AP PRECALCULUS

Unit 1 • Practice

# Polynomial and Rational Functions

AP-style practice • Multiple choice and free response

**About this practice set**

The questions follow the format of the AP Precalculus exam. Part A and Part B are multiple choice; Part C is free response. Each question is labeled **NO CALCULATOR** or **CALCULATOR**. Suggested timing: Part A 35 min • Part B 10 min • Part C 45 min.

## Part A • Multiple choice • No calculator

**Directions • Part A**

A calculator may **not** be used on this part. Solve each problem, then choose the best of the four answer choices. Unless otherwise stated, the domain of a function $f$ is the set of all real numbers $x$ for which $f(x)$ is a real number. Some questions refer to a graph: read the values you need from the grid.

### Question 1 (1.1)

NO CALCULATOR

The table gives values of a function $f$ at selected values of $x$.

| $x$ | 0 | 2 | 4 | 6 |
|---|---|---|---|---|
| $f(x)$ | 5 | 9 | 11 | 12 |

Which statement is consistent with the table?

(A) $f$ is increasing and its graph is concave up.

(B) $f$ is increasing and its graph is concave down.

(C) $f$ is decreasing and its graph is concave up.

(D) $f$ is decreasing and its graph is concave down.

### Question 2 (1.2)

NO CALCULATOR

What is the average rate of change of $f(x) = x^3 - 2x$ on the interval $[-1, 2]$?

(A) $3$

(B) $\frac{5}{3}$

(C) $1$

(D) $-1$

### Question 3 (1.3)

NO CALCULATOR

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | 2 | 5 | 10 | 17 | 26 |

Which type of function best models the data in the table?

(A) Quadratic

(B) Linear

(C) Cubic

(D) Constant

**Question 4** (1.5) NO CALCULATOR

A polynomial $p$ of degree 4 has real coefficients. Two of its zeros are 2 and $1+3i$. Which of the following must also be a zero of $p$?

(A) $-2$
(B) $3+i$
(C) $-1-3i$
(D) $1-3i$

**Question 5** (1.6) NO CALCULATOR

Let $p(x) = x^2(x-3)^3(x+1)$. What is $\lim_{x\to-\infty} p(x)$?

(A) $-\infty$
(B) $+\infty$
(C) $0$
(D) $1$

**Question 6** (1.5) NO CALCULATOR

Which statement describes the graph of $p(x) = (x+2)^2(x-1)$ at $x=-2$?

(A) It has a vertical asymptote.
(B) It crosses the $x$-axis.
(C) It touches the $x$-axis without crossing it.
(D) It has a hole.

**Question 7** (1.7) NO CALCULATOR

Which line is a horizontal asymptote of the graph of $r(x) = \dfrac{3x^2-x}{6x^2+5}$?

(A) $y=3$
(B) $y=0$
(C) $y=-\dfrac{1}{5}$
(D) $y=\dfrac{1}{2}$

**Question 8** (1.10) NO CALCULATOR

Let $r(x) = \dfrac{x^2-1}{x^2+x-2}$. Which statement is true about the graph of $r$ at $x=1$?

(A) It has a vertical asymptote $x=1$.
(B) It has a hole at $\left(1, \frac{2}{3}\right)$.
(C) It has a zero at $x=1$.
(D) It has a hole at $(1,0)$.

**Question 9** (1.9) NO CALCULATOR

What is $\lim_{x\to2^+} \dfrac{x+1}{2-x}$?

(A) 0
(B) 3
(C) $+\infty$
(D) $-\infty$

**Question 10** (1.11) NO CALCULATOR

The graph of $r(x) = \dfrac{2x^2+3x-4}{x+2}$ has a slant asymptote. What is its equation?

(A) $y = 2x - 1$
(B) $y = 2x + 7$
(C) $y = 2x + 3$
(D) $y = x - 1$

**Question 11** (1.11) NO CALCULATOR

What is the coefficient of $x^2$ in the expansion of $(x-3)^4$?

(A) 6
(B) $-108$
(C) 54
(D) 9

**Question 12** (1.12) NO CALCULATOR

The function $f$ has domain $[0,8]$ and range $[-2,3]$. What is the range of $g(x) = -2f\left(\frac{x}{2}\right) + 1$?

(A) $[-5,5]$
(B) $[-3,7]$
(C) $[-4,6]$
(D) $[-1,3]$

**Question 13** (1.5) NO CALCULATOR

Which function is even?

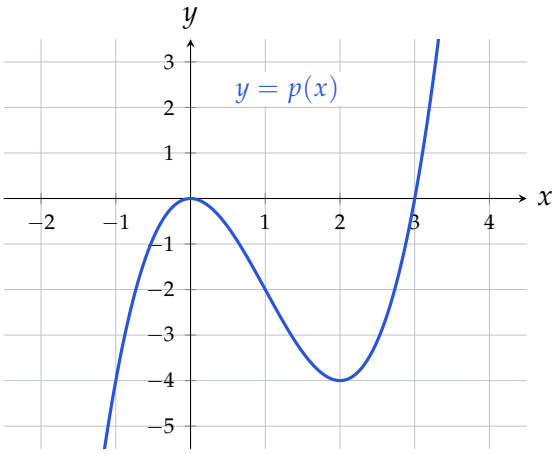
(A) $f(x) = x^3 - x$
(B) $f(x) = x^4 + 3x^2 - 1$
(C) $f(x) = x^2 + x$
(D) $f(x) = (x-1)^2$

**Question 14** (1.8) NO CALCULATOR

What is the solution set of $\dfrac{x-3}{x+1} \geq 0$?

(A) $(-\infty,-1] \cup [3,\infty)$
(B) $[3,\infty)$
(C) $(-1,3]$
(D) $(-\infty,-1) \cup [3,\infty)$

**Questions 15 to 18 refer to the graph of the polynomial $p$ below.**




**Question 15** (1.5 • graph) NO CALCULATOR

Which of the following could be an expression for $p(x)$?

(A) $x(x-3)^2$
(B) $-x^2(x-3)$
(C) $x^2(x-3)$
(D) $x^2(x+3)$

**Question 16** (1.4 • graph) NO CALCULATOR

On which interval is $p$ decreasing?

(A) $(-\infty, 0)$
(B) $(0, 2)$
(C) $(2, \infty)$
(D) $(0, 3)$

**Question 17** (1.2 • graph) NO CALCULATOR

What is the average rate of change of $p$ on the interval $[1, 3]$?

(A) $-2$
(B) $0$
(C) $2$
(D) $1$

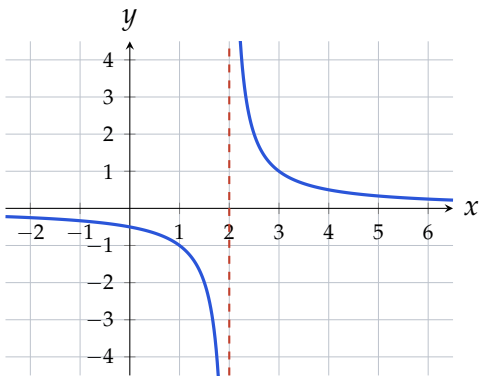
**Question 18** (1.4 • graph) NO CALCULATOR

The graph of $p$ has a point of inflection at $(1, -2)$. On which interval is the graph of $p$ concave up?

(A) $(1, \infty)$
(B) $(-\infty, 1)$
(C) $(0, 2)$
(D) $(-\infty, 0)$

**Question 19** (1.9 • graph) NO CALCULATOR

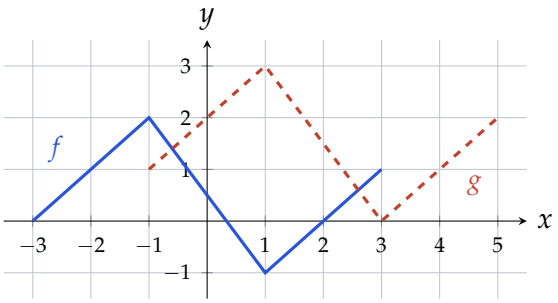
The graph of a rational function $s$ is shown. It has a vertical asymptote $x = 2$ and a horizontal asymptote $y = 0$, and it passes through $(3, 1)$ and $(1, -1)$. Which could be an expression for $s(x)$?




(A) $\dfrac{1}{x+2}$
(B) $\dfrac{1}{x-2}$
(C) $\dfrac{x}{x-2}$
(D) $\dfrac{x-2}{x}$

**Question 20** (1.12 • graph) NO CALCULATOR

The graphs of $f$ (solid) and $g$ (dashed) are shown. Which is an equation for $g$ in terms of $f$?




(A) $g(x) = f(x-2)+1$
(B) $g(x) = f(x+2)+1$
(C) $g(x) = f(x-2)-1$
(D) $g(x) = f(x+2)-1$

## Part B • Multiple choice • Graphing calculator

**Directions • Part B**

A graphing calculator is **required** for the questions in this part. If the exact value of the correct answer is not one of the choices, choose the one that best approximates it.

**Question 21** (1.14) CALCULATOR

An open box is made from a 16 in by 10 in sheet by cutting squares of side $x$ in from the corners. Its volume is $V(x) = x(16-2x)(10-2x)$ for $0 < x < 5$. What is the maximum volume of the box?

(A) 120 $\text{in}^3$
(B) 144 $\text{in}^3$
(C) 160 $\text{in}^3$
(D) 168 $\text{in}^3$

**Question 22** (1.14) CALCULATOR

The height $h$, in meters, of a model rocket was measured at times $t$, in seconds.

| $t$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $h(t)$ | 1.2 | 6.1 | 9.0 | 9.9 | 8.6 |

Using a quadratic regression, what is the predicted height at $t = 5$ seconds?

(A) 4.2 m
(B) 5.3 m
(C) 6.4 m
(D) 7.3 m

## Part C • Free response

**Directions • Part C**

Show all of your work: your answers are graded on the correctness and completeness of your methods as well as on the final answers. When a question asks you to justify or explain, give mathematical reasons in words. Use a calculator only on the question labeled CALCULATOR. Decimal answers should be correct to three places after the decimal point.

**Question 23** (Function concepts) NO CALCULATOR

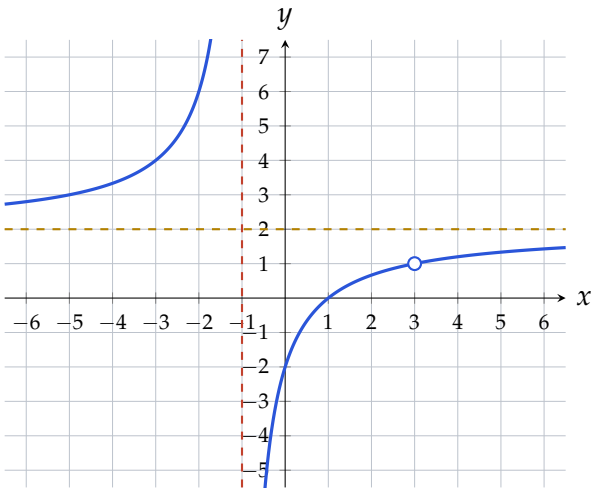
The polynomial function $h$ has degree 3 and leading coefficient 1. Selected values of $h$ are given in the table.

| $x$ | $-2$ | $-1$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|
| $h(x)$ | $-1$ | 3 | 1 | $-1$ | 3 | 19 |

(A) (i) Find the average rate of change of $h$ over the interval $[-1, 1]$. Show the work that leads to your answer.
(ii) How many real zeros does $h$ have? Give a reason for your answer using the table.
(B) (i) The average rates of change of $h$ over $[1, 2]$ and $[2, 3]$ are different. Use them to determine whether the graph of $h$ is concave up or concave down on $[1, 3]$. Give a reason for your answer.
(ii) Write a limit statement that describes the end behavior of $h$ as $x$ decreases without bound.
(C) Let $k(x) = h(x - 2) + 4$. Find $k(3)$, and describe how the graph of $k$ is obtained from the graph of $h$.

**Question 24** (Graph analysis) NO CALCULATOR

The graph of the rational function $r(x) = \dfrac{ax^2 + bx + 6}{x^2 - 2x - 3}$, where $a$ and $b$ are constants, is shown. The graph has a vertical asymptote $x = -1$, a horizontal asymptote $y = 2$, an $x$-intercept at $(1, 0)$, and a hole at $(3, 1)$.




(A) (i) Write limit statements that describe the behavior of $r$ near $x = -1$, using the graph.
(ii) Write a limit statement that describes the end behavior of $r$.

(B) Use the horizontal asymptote and the $x$-intercept to find the values of $a$ and $b$. Show that your expression has a hole at $(3, 1)$.

(C) Using the graph, solve $r(x) < 0$. Express your answer in interval notation.

**Question 25** (Modeling a non-periodic context) CALCULATOR

A theater studies how its revenue $R$, in dollars, depends on the ticket price $p$, in dollars.

| $p$ (dollars) | 10 | 15 | 20 | 25 | 30 |
|---|---|---|---|---|---|
| $R$ (dollars) | 4000 | 5250 | 6000 | 6250 | 6000 |

(A) (i) Explain why a quadratic function is an appropriate model for these data.
(ii) Write an expression for a quadratic function $R(p)$ that fits the data exactly.

(B) (i) Use your model to predict the revenue when the ticket price is \$35.
(ii) Find the average rate of change of $R$ from $p = 25$ to $p = 35$, and interpret its meaning in context, with units.

(C) Revenue cannot be negative. Find the interval of ticket prices for which the model gives a revenue of at least \$0, and state one reason the model may not be valid for every price in that interval.

**Question 26** (Symbolic manipulations) NO CALCULATOR

Let $r(x) = \dfrac{2x^2 + 2x - 4}{x^2 - 1}$ and $p(x) = x^3 - 2x^2 - 5x + 6$.

(A) (i) Find the zeros of $r$ and the coordinates of any hole in the graph of $r$.
(ii) Find the vertical asymptote of the graph of $r$. Write limit statements that justify your answer.

(B) (i) Write a limit statement that describes the end behavior of $r$.
(ii) Solve $r(x) \geq 0$. Express your answer in interval notation.

(C) It is known that $p(1) = 0$. Write $p(x)$ as a product of linear factors, and find all zeros of $p$.

Check your answers with the answer key for this unit.